Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/02/2022
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	GeneLab: Extending the Reach of the GeneLab Data System Across Kingdoms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	12/17/2018	**End Date:**	12/16/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/16/2021 per NSSC information (Ed., 12/30/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin System)		
Grant/Contract No.:	80NSSC19K0132		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The GeneLab data repository is a rapidly expanding and remarkably powerful resource with which to dissect organismal responses to spaceflight at levels ranging from the metagenome to the transcriptome and proteome. The research in this GeneLab Data System study sought to extend the bioinformatics tool development for GeneLab data to a cross-kingdom architecture, making inter-organismal comparisons possible. This approach facilitates mining across the wealth of datasets within the GeneLab system and drives insights into the fundamental responses to spaceflight shared by biological systems as a whole. This work has approached some of GeneLab's core questions as to how implementation of large-scale dataset analyses can help produce new insights and hypotheses about the effects of spaceflight on biology. . An interactive data visualization and mining environment has been developed based on the Qlik engine, which covers all the biology data deposited within GeneLab. This approach allows users to make interactive comparisons and filters between datasets to mine for both common and unique features of the transcriptomes of the GeneLab data. Current work is focused on maintaining and expanding this data environment and moving towards a highly intuitive user interface. This tool will allow users to rapidly move from data exploration to identifying spaceflight-response genes and from there to testable hypotheses. Specific aims: (1) Import the diversity of the datasets from the GeneLab Data repository into the Qlik database and develop Orthologous Matrix- and Gene Ontology-driven approaches to translate between them in a kingdom-transparent manner. (2) Explore using common phenotyping tools and approaches for the research community to facilitate the testing of potential roles of these genes in spaceflight-related factors. This effort will focus on elements of spaceflight such as microgravity and hypoxic stress response that are applicable to a wide range of organisms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has been to develop an in silico analysis environment to explore GeneLab's rich set of spaceflight and spaceflight-related data. This approach should help uncover not only patterns of 'omic-level responses to spaceflight but also provide insight into how biology reacts to the range of stimuli related to spaceflight such as to gravity, radiation, and mechanical forces on Earth. The database allows comparisons across Kingdoms, providing a new tool for biologists to also ask how universal the molecular responses they are characterizing are across the diversity of life on Earth.
Task Progress:	I. Completion of the TOAST database: The Test Of Arabidopsis Space Transcriptome (TOAST) database is a web-based data visualization and analysis environment for all publicly available plant science spaceflight data. The system imports data from GeneLab, from the original publications on each study and from datasets made available by the primary researchers. These data have then been manually curated and linked in a relational database that allows researchers to rapidly compare transcriptome patterns across studies of interest. This tool has been through extensive beta testing and optimization and the fully functional version is publicly available at: https:// The paper describing the TOAST data visualization environment was published in 2020 in Frontiers in Plant Science. II. Ortholog mapping using the Cross-Species Explorer: A major goal for this research program has been to extend the data visualizations outlined above for TOAST to all the data available in the GeneLab data repository, including animal, plant, and microbial datasets. To this end, equivalent genes (orthologs) have been mapped across species. All the datasets deposited in GeneLab have then been imported and visualization and cross-referencing tools developed to make these comparisons transparent to the user. The web app to access this data engine (called the X-species Explorer) is publicly available at: https:// III. The Rad-Bio-App: In a parallel effort to help space biologists explore the effects of radiation across the available spaceflight omics-level data, a data visualization environment called the Rad-Bio-App has been developed. All the available radiation dosage data from the spaceflight experimentation deposited in GeneLab, as well as that from ground-based studies that have used a variety of sources and accelerators, have been imported and curated alongside associated metadata ranging from the source and type of radiation to species analyzed and hardware used to perform the experiment. All these data are then linked in a database to the associated Gene Lab data accessions, along with available data from the lunar and Martian environments. The App then allows users to explore the GeneLab datasets for their associated radiation environments and filter for similar and different experiment parameters. RadBioApp has 2 goals: (1) providing a useful tool for space radiation biologists, and (2) making analysis of radiation effects more accessible to the non-specialist researcher. The tool is available at: https:// IV. Metadata analyses -The Matrix: GeneLab has developed standardized pipelines for analysis of microarray and RNAseq to help make data comparisons between studies more robust. In collaboration with the GeneLab Plant Analysis Working Group (Plant AWG), a curated metadata matrix (The Matrix) that cross-references to these GeneLab analyses has been developed. The combination of The Matrix and the standardized analyses allows researchers to ask what variables affect the spaceflight data most significantly. This tool is available at: https:// V. Presentations, Outreach, and Education: During the course of this funding, the data visualizations made possible by this research have allowed presentations to be made at the annual meetings of the American Society for Gravitational and Space Research, the Space Generation Advisory Council in Taiwan, The International Space Station Research and Development (ISS R&D) Conference, the annual meeting of the American Society of Plant Biologists, the Midwest Plant Cell Dynamics Meeting, NASA Kennedy Space Center, the Academia Sinica in Taiwan, and at Carthage College. This space research has also formed the basis for outreach efforts at events ranging from University of Wisconsin sponsored outreach days (e.g., University of Wisconsin's Science Expeditions), to work with groups such as the University of Wisconsin's PEOPLE program (that engages under-represented groups in science), to Madison's BioHouse (that engages with high-achieving undergraduates), to middle school and K-12 teachers (the BioPharmaceutical Technology Center Institute training program), to the Kids in Data program designed to introduce young students to concepts of data analysis, to space camp for 5th grades at the Deke Slayton Bike and Space Museum in Sparta, WI.

	15 independent study students have worked in the Gilroy lab most semesters on various plant molecular and engineering projects related to space biology, with a team of ~4 contributing to user interface design for the GeneLab X-Species Explorer. Four undergraduates and one high school student have also been engaged in bioinformatics-driven projects related to GeneLab data analysis and visualization over this period.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
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